KUKA

ROBOT

(V)KR 6; (V)KR 16; ...-2

Maintenance

Manipulator without Wrist

Issued: 20 Mar 2013 **Version: 08**

© Copyright 2013

KUKA Roboter GmbH
Zugspitzstraße 140
D-86165 Augsburg

This documentation or excerpts therefrom may not be reproduced or disclosed to third parties without the express permission of the publishers.
Other functions not described in this documentation may be operable in the controller. The user has no claim to these functions, however, in the case of a replacement or service work.
We have checked the content of this documentation for conformity with the hardware and software described. Nevertheless, discrepancies cannot be precluded, for which reason we are not able to guarantee total conformity. The information in this documentation is checked on a regular basis, however, and necessary corrections will be incorporated in subsequent editions.
Subject to technical alterations without an effect on the function.

Contents

1 General 5

2 Lubrication work 7

2.1 Changing the oil on main axis gear units 10
2.1.1 Changing the oil on axis 1 gear unit 10
2.1.2 Changing the oil on axis 2 gear unit 11
2.1.3 Changing the oil on axis 3 gear unit 13

3 Other maintenance 15

4 Cleaning and preventive maintenance 17

5 Safety precautions when handling lubricants 18

Valid for **(V)KR 6; ...-2**

(V)KR 16; ...-2 **(V)KR 16 EX; ...-2 EX** **(V)KR 16 CR; ...-2 CR**
(V)KR 16 S; ...-2 S **(V)KR 16 L6; ...-2**

1 General



This description applies analogously to all of the robots listed above, regardless of the robot variant depicted.



The Doc. Module “Safety, General” is to be observed!

With “EX” variant robots, work may only be carried out if there is no risk of explosion.

Maintenance and cleaning work on CR-variant robots may only be conducted outside cleanrooms. Instructions of the cleanroom user must be observed. Cleaning operations are to be carried out with cleaning agents suitable for use in the cleanroom.

Before any maintenance work is started, any attached tools or additional equipment that would hinder maintenance must be dismounted.

Lubrication is performed either at the specified maintenance intervals or every 5 years after commissioning by the customer – whichever is reached first.

With maintenance intervals of 20 000 hours, the first maintenance (oil change) is performed either after 20 000 operating hours or 5 years after commissioning by the customer – whichever is reached first.



It is imperative that lubrication is carried out every 5 years, as prescribed (oil ageing).



The maintenance intervals given in the tables are valid for the operating conditions specified in the robot Doc. Module “Technical Data”. In case of variations from normal conditions (e. g. increased dust or water content in the environment of the robot, abnormal temperatures), the KUKA design department must be consulted.

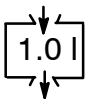
Explanation of symbols:



Maintenance point


Maintenance point, visible on drawing


Maintenance point, hidden on drawing



Oil change, to be repeated at specific intervals;

with indication of oil capacity


Tighten screw (nut)


Check and replace toothed belts


Visual inspection

The description of maintenance operations is subdivided into job steps with numbers in brackets appearing before them. The text which immediately follows these steps must also be read if it is specially marked by a **warning triangle** or either of the **hand symbols**. Many of these marked texts refer to the preceding job step.



Example:

(1) Pour specified amount of oil into filler hole (10).



Only KUKA-approved lubricants may be used.

Some of the specially marked texts refer exclusively to the subsequent steps – until the instruction is expressly revoked or the work is completed at the end of a section.



Example:



Turn main switch on the robot control cabinet to "OFF" and secure it with a padlock to prevent unauthorized persons from switching it on again.

2 Lubrication work



Chapter 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.



Utmost caution must be exercised when handling cable grease. The use of protective clothing (at least protective gloves, for example) is strongly recommended.
All accident prevention regulations and Section 5 of this Doc. Module must be observed.



Store used oil and grease in accordance with regulations and dispose of them with minimum environmental impact.

- **“Lubrication” table**

Maintenance point (Fig. 1)	maintenance interval (hours) 10000	maintenance interval (hours) 20000	Activity Lubricant Manufacturer Amount KUKA Art. no.	Comment
① **Floor-mounted manipulator** Gear with bearings A 1	● 2)	●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.500 l 1) KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.1.
① **Ceiling-mounted manipulator** Gear with bearings A 1	● 2)	●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.600 l 1)3) KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.1.
② Gear with bearings A 2	● 2)	●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.500 l 1) KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.2.
③ Gear with bearings A 3	● 2)	●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.200 l 1) KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.3.

● Normal maintenance interval
1) The oil quantities specified are the actual amounts of oil in the gear unit at first filling
2) For EX-variant robots
3) 0,600 l only with compensation vessel

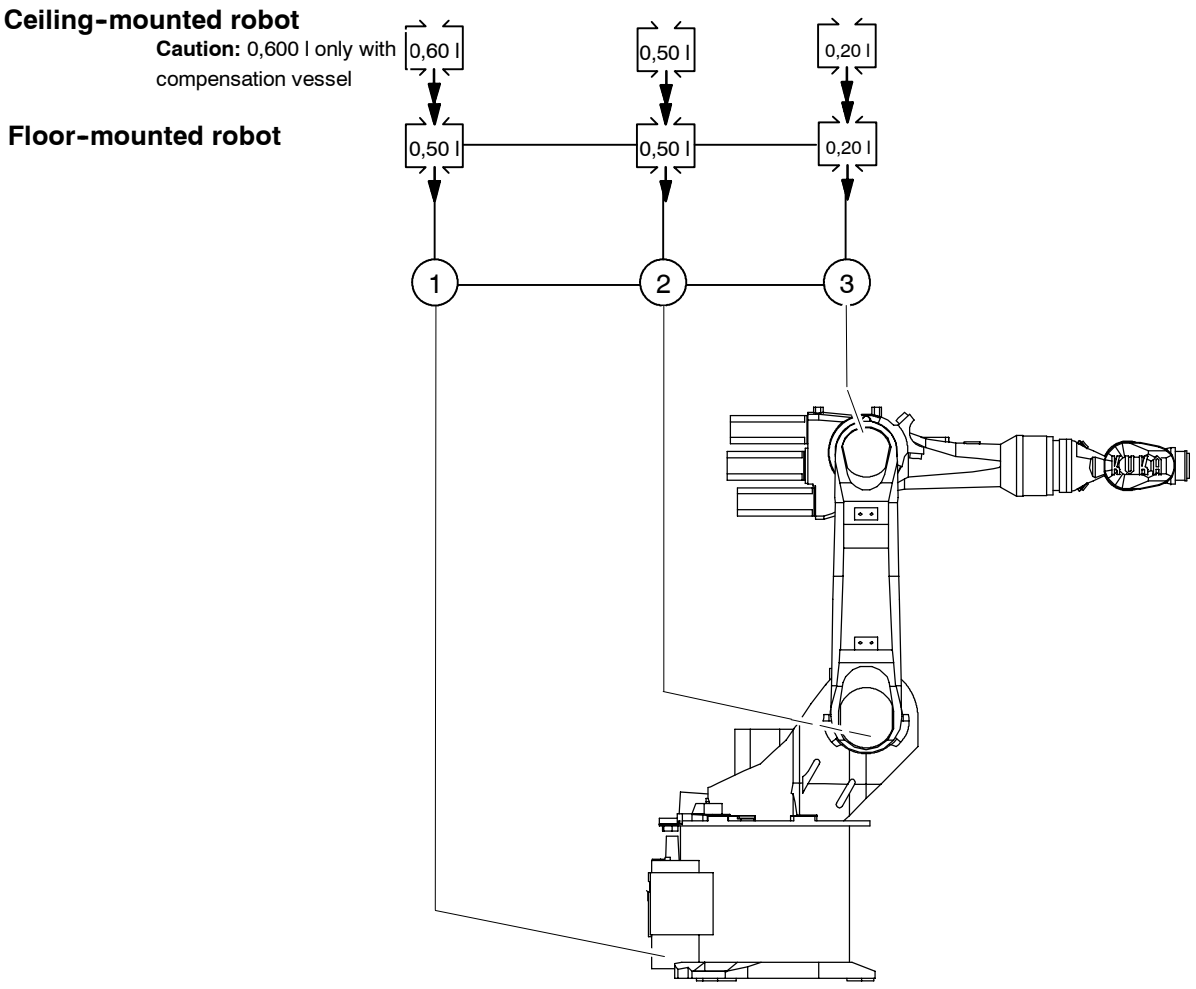


Fig. 1 Lubrication

When draining the oil, it is important to remember that the quantity drained is dependent on time and temperature.
The quantity of oil drained must be measured, as only this quantity may be used when refilling. If less than 70% of the specified oil quantity flows out, flush the gear unit with the measured quantity of drained oil once, then pour in the amount of oil that was drained. During the flushing procedure, move the axis at jog velocity throughout the entire axis range.

2.1 Changing the oil on main axis gear units

Chapter 2 of this Doc. Module is to be observed!

The oil is to be changed only at operating temperature.

If the oil change is carried out immediately after the robot has stopped operating, the oil temperature is liable to be high, in which case appropriate safety measures must be taken.

2.1.1 Changing the oil on axis 1 gear unit

Section 2.1 of this Doc. Module is to be observed!

- **Draining the oil**

(1) Remove four M6x10 Allen screws (Fig. 2/4) and lock washers, and take off cover (5).

(2) Remove cable straps (3) and pull out drain hose (6).

(3) Unscrew sealing cap (7) on drain hose (6) and place receptacle (10) under the drain hole.

(4) Remove magnetic screw (8) and check magnet for swarf or metal particles.

(5) Catch the oil as it drains out

The oil takes about 15 minutes to drain completely.

Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

(6) Screw sealing cap (7) onto drain hose (6) and tighten it.

- **Filling with oil**

(1) Pour specified amount of oil into filler hole of magnetic screw (8).

Use a hose (2) approx. 1 m long with a straight M18x1.5 union (9) together with a funnel (1) which fits into the free end of the hose (2).

(2) Remove filling hose (2).

(3) Insert and tighten magnetic screw (8), tightening torque M_A = 20 Nm.

(4) Check sealing cap (7) for leaks.

(5) Attach cable straps (3).

(6) Mount cover (5) and fasten it with four M6x10 Allen screws (4) and lock washers.

(7) Connect any peripheral supply lines required.

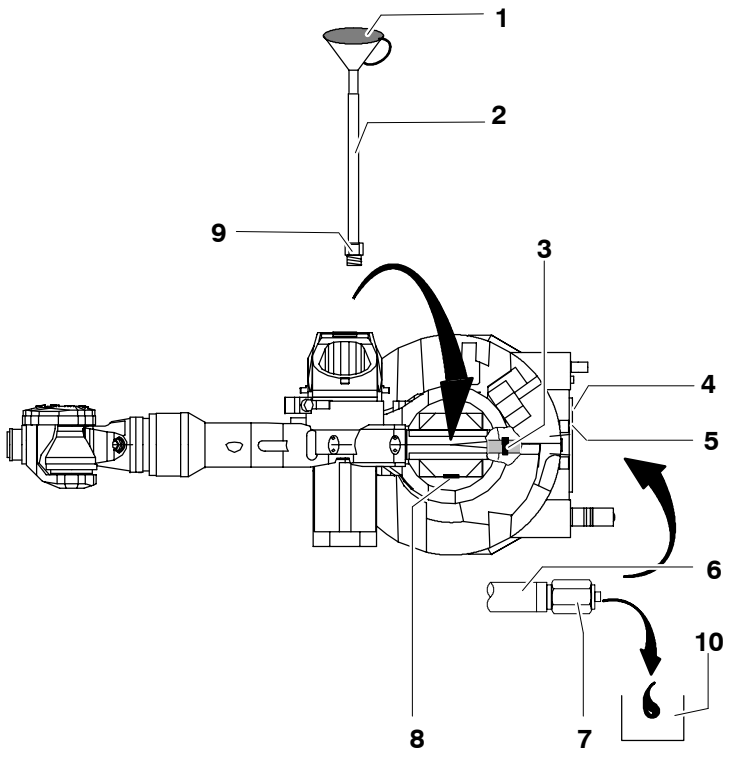


Fig. 2 Changing the oil on axis 1

2.1.2 Changing the oil on axis 2 gear unit

Section 2.1 of this Doc. Module is to be observed!

For oil draining and filling use a hose (Fig. 3/2) approx. 1 m long with a straight M18x1.5 union (3) together with a funnel (1) which fits into the free end of the hose (2).

- **Drain oil**

(1) Remove screw plug (5).

(2) Screw in drain hose and place receptacle (6) under the hose.

(3) Remove screw plug (4) and catch oil as it drains out.

The oil takes about 15 minutes to drain completely.

Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

(4) Remove drain hose.

(5) Insert and tighten screw plug (5); tightening torque M_A = 20 Nm.

- **Fill with oil**

(1) Pour specified amount of oil into the filler hole at (4).

Only KUKA-approved lubricants may be used.

(2) Remove filling hose.

(3) Insert and tighten screw plug (4); M_A = 20 Nm.

(4) Check screw plug (5) for leaks.

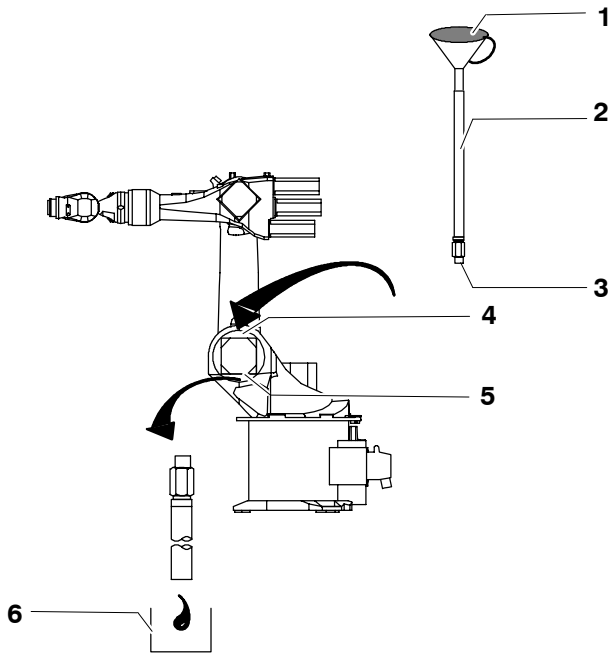


Fig. 3 Changing the oil on axis 2

2.1.3 Changing the oil on axis 3 gear unit



Section 2.1 of this Doc. Module is to be observed!

- **Drain oil**



Before performing the next step, it must be ensured that it is not possible for anyone to be injured within the range of the slowly turning robot.
The robot may only be moved at jog speed, with all applicable safety rules and regulations being observed.

(1) Put robot into operation and move axis 3 until the magnetic screws (Fig. 4/2, 6) are aligned vertically.



Turn main switch on the robot control cabinet to "OFF" and secure it with a padlock to prevent unauthorized persons from switching it on again.

(2) Remove magnetic screw (6) and place receptacle (5) under the drain hole.



It is easier to drain the oil if a hose (4) with a straight M10x1 union (3) is used.

(3) Remove magnetic screw (2) and catch oil as it drains out.



The oil takes about 10 minutes to drain completely.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

(4) Remove drain hose.

(5) Insert and tighten magnetic screw (6) (tightening torque M_A = 10 Nm).

- **Fill with oil**

(1) Pour specified amount of oil into the filler hole at (2).

Only KUKA-approved lubricants may be used.

It is easier to pour in the oil if a funnel (1) is used for this purpose.

(2) Insert and tighten magnetic screw (2); M_A = 10 Nm.

(3) Check magnetic screws (2, 6) for leaks.

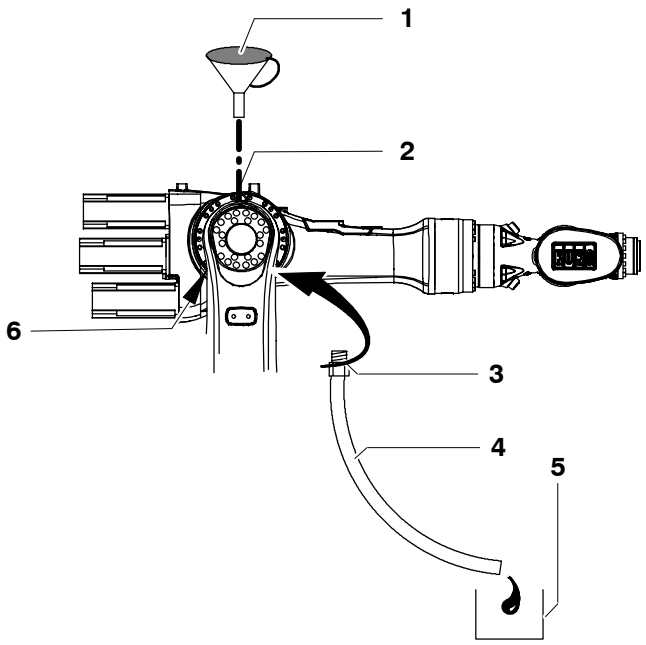


Fig. 4 Changing the oil on axis 3

3 Other maintenance



Chapter 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

- **“Other maintenance” table**

Maintenance point (Fig. 5)		maintenance interval (hours) 10000	Activity	Comment
①	Robot installation	●	Check tightening torques for bolts and anchor nuts.	See Doc. Module Robot, ”Installation, Connection, Exchange”.
②	Electric cables and dress package	● 2)	Check electric cables, flexible tubes and dress package for wear and tear.	Exchange any damaged cables and flexible tubes. See robot Doc. Module, “Repair, Electrical Installations”.
		● 2)	Check the quantity of cable grease in the base frame and replenish if necessary. Cable grease Art. no. 00-126-265	
③	Toothed belt, arm A 4, A 5	● 1)	Check toothed belt tension Checking toothed belts	See robot Doc. Module “Adjustment Instructions” Replace toothed belts if cracks or signs of deterioration are found. See robot Doc. Module, ”Repair, Motor Units”.
		● 3)	Replace toothed belts	To exchange the toothed belts, see robot Doc. Module, “Repair, In-line Wrist”.

1) Carried out at KUKA before delivery, or must be performed by the user after a toothed belt has been exchanged.
2) For EX-variant robots, after 5,000 hours of operation.
3) After 2 years at the latest.

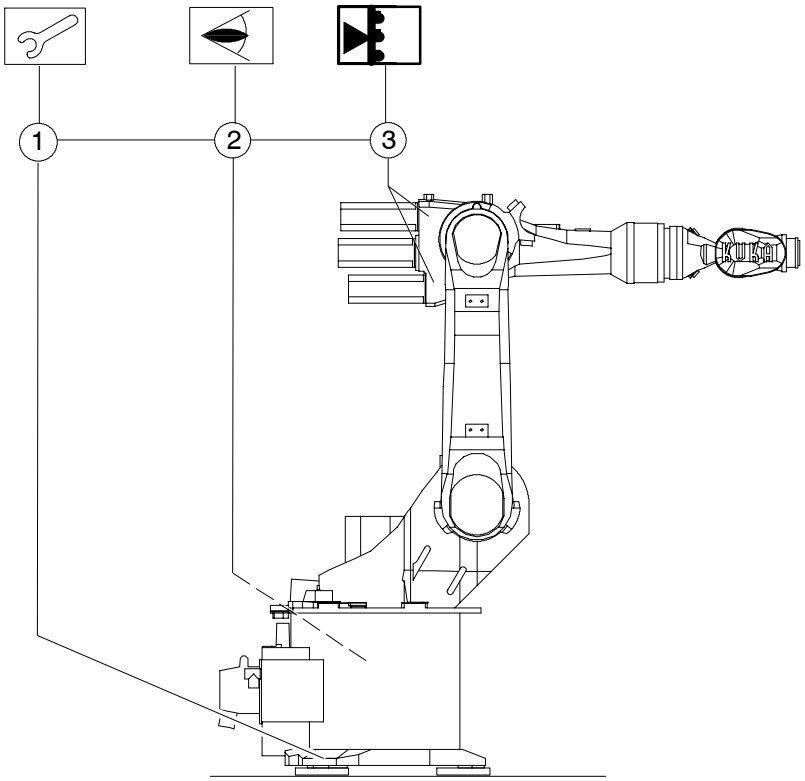


Fig. 5 Other maintenance

4 Cleaning and preventive maintenance



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again. This also applies to any cleaning work, which though not performed directly on the robot is carried out within or close to its working range.

With “EX” variant robots, work may only be carried out if there is no risk of explosion.

Maintenance and cleaning work on CR-variant robots may only be conducted outside cleanrooms. Instructions of the cleanroom user must be observed. Cleaning operations are to be carried out with cleaning agents suitable for use in the cleanroom.

Cleaning and preventive maintenance work is to be carried out in accordance with the following instructions:

- Clean robot with a cloth soaked with washing agent.
- Clean cables, painted surfaces, plastic parts and hoses with solvent-free washing agents.



The manufacturer’s instructions are to be observed when using washing agents.

- Remove any lubricant that has escaped using washing agent. If a large amount of lubricant has emerged, trace and remedy the cause.



It must be ensured that no washing fluid enters bearings, seals or electrical equipment.



If washing agent does enter bearings, these must be re-greased or re-oiled.

- Remove any corrosion and – where permissible – protect the affected areas with paint, grease or oil.
- Apply a thin coat of oil to bare metal parts.
- Replace damaged, illegible or missing inscriptions, labels and plates.



Do not use compressed air to clean the system, as this could cause dirt to penetrate and damage seals, bearings and electrical components.
Used cleaning agents and lubricants must be stored and disposed of in accordance with all pertinent regulations and with minimum environmental impact.

5 Safety precautions when handling lubricants

The information contained in the safety data sheet according to 91/155 EEC must be observed when handling lubricants. An excerpt from this data sheet is included in the robot Doc. Module “Consumables, Safety Data Sheet”.

The following safety procedures must be observed at all times:

- Avoid prolonged intensive skin contact; wear protective gloves and aprons if necessary (especially when working with cable grease).



All accident prevention regulations must be observed when handling lubricants.

- If lubricant is likely to come into contact with the skin, e.g. with the hands, suitable barrier creams must be applied before work is started.
- On completion of work and before smoking breaks or meal breaks (before any intake of food), clean oil-contaminated skin thoroughly with water and skin-friendly cleansing agents or soap. After cleaning, replace lost skin grease with appropriate skin creams containing grease.
- Change oil-soaked clothing immediately. Do not carry oily rags or any cloths soaked with solvents, mineral oil mixtures or cable grease in your pockets.
- Avoid breathing in oil spray or vapors wherever possible.